

IMPACT OF NIRSEVIMAB ON PEDIATRIC RSV HOSPITALIZATIONS

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Sciensano, Epidemiology of Infectious Diseases
Thanks to Sophie Blumental

INTRODUCTION

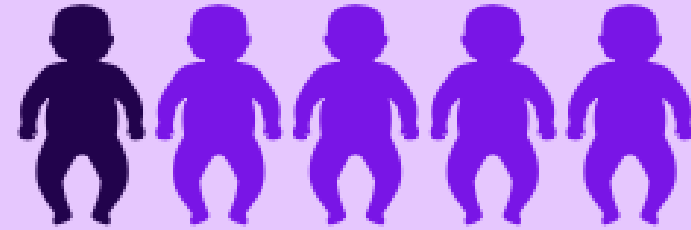
Disclosure

Publication in progress
Confidentiel data

RSV burden of disease (1)

RSV is the leading cause of **lower** respiratory tract infections in children **younger than 5 years old**

Outpatient Visits



1 in 5

children will need outpatient care **by age 2** due to RSV¹

RSV burden of disease (1)

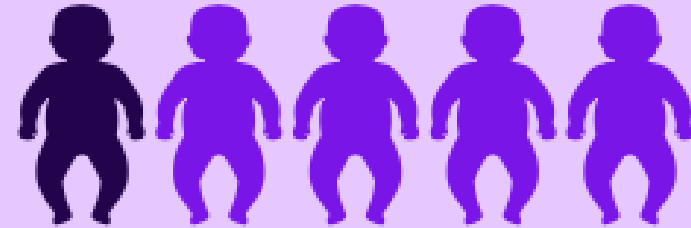
almost all children aged < 2 years old experience at least one episode of RSV infection

Approximately 40% of all primary RSV infections in infancy result in LRTI, principally bronchiolitis and pneumonia
RSV is the leading cause of **lower** respiratory tract infections in children **younger than 5 years old**

12. Paes BA, Mitchell I, Banerji A, Lanctôt KL, Langley JM., A decade of respiratory syncytial virus epidemiology and prophylaxis: translating evidence into everyday clinical practice. <https://www.mdpi.com/2076-2607/8/12/2048>

 <https://respiratoryjournal.be/2019/8/12/2048>
Canadians, the burden of RSV, how the scientific community can help/
Confidential data manuscript in progress

Outpatient Visits



1 in 5

children will need outpatient care by age 2 due to RSV¹

1. Lively JY, et al. *J Pediatric Infect Dis Soc.* 2019;8(3):284-286

RSV burden of disease (2)

*Number of Hospitalization due to a proven RSV infection in **Belgium** :*

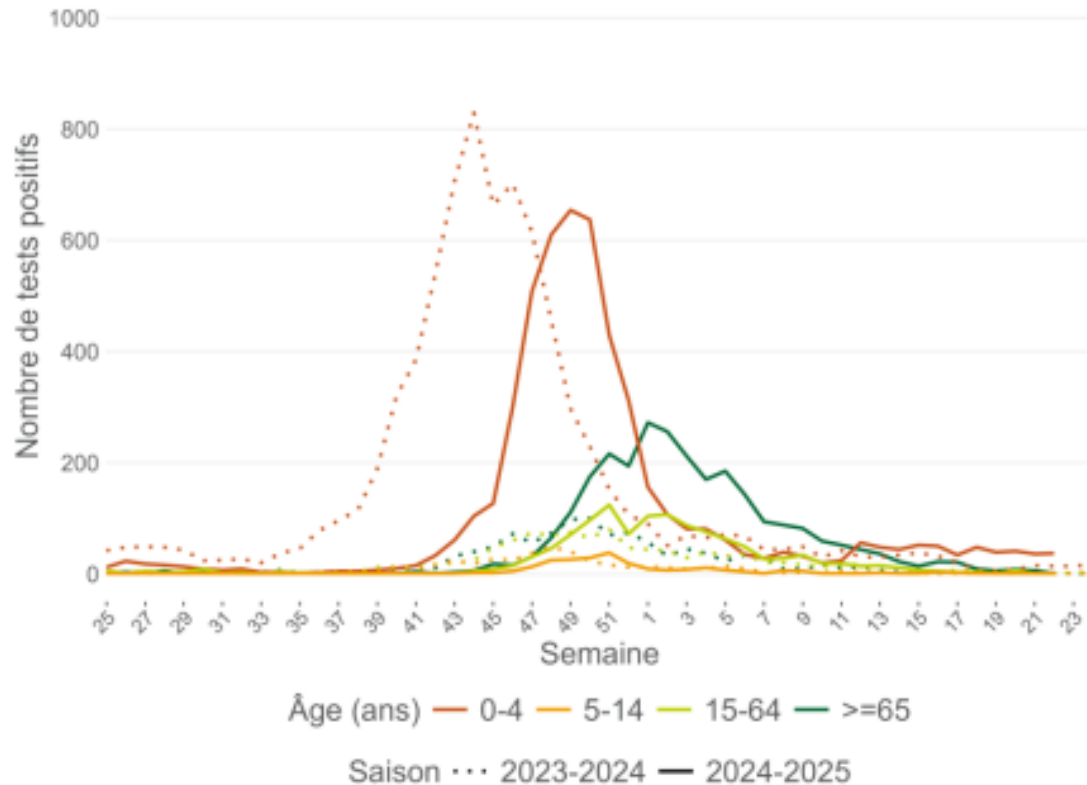
More than **10.000** children under 5 years (2%)

More than **7.000** children under the age of 1 year (7%)



Bouckaert N et al. *Pediatr Infect Dis J* 2023; 42(10): 857-861

Seasonality of RSV in Belgium



Children under 5 years old are the most impacted

Seasonality pattern

Single peak between **October** and **January**

Narrow peak

Increase pressure in health care system during this period

New preventive tools against RSV in children available from 2024

Nirsevimab (Beyfortus)

Long-acting monoclonal anti-RSV **antibodies**
Reimbursed Belgium **June 2024**



Abrysvo

Maternal RSV Pre-F **vaccine**
Reimbursed Belgium: **January 2025**



Nirsevimab eligible population in 24-25

Infants, under the age of 6 months, facing their first potential RSV exposure season were eligible for nirsevimab:

- **For babies born during the RSV season** : administration at birth
- **For the others** : catch-up administration just before beginning of the RSV season

In 2023–2024 **69%** of hospitalized children below the age of 5 would have been eligible

DATA COLLECTION

SARI survey

- Prospective inclusion of all cases hospitalized for severe acute respiratory infections matching the case definition
- Sample sent to the reference Lab, identification of respiratory viruses by pcr
- Weekly clinical data report by participating centers (pseudonymized data)
- Weekly bulletin of acute respiratory infections

RSVPED study

- Active recruitment of 25 centers in Belgium outside the SARI network
- Inclusion of **all proven RSV** cases **0-4 years old** meeting the case definition and hospitalized during the enlarged epidemic period (October to March)
- Retrospective **clinical data collection** by local team/data nurse (short or long CRF) (pseudonymized data for season 2023-2024)

RESULTS

Estimation of coverage

SARI data

Estimation based on the proportion of eligible in 23-24, in 24-25 and IE):

74%

Reimbursement approval data (IMA)

62.5%

Pilot study

led by Marc Raes (lead of the RSV task force):

80-85% *(born during the season)* **54%** *(born before the season)*

Estimation of immunization effectiveness

Test-negative design (common in vaccine studies):

Compare immunisation status between

- a) Nirsevimab-eligible SARI patients hospitalized because of RSV (**cases**)
- b) to those hospitalized for SARI but tested negative for RSV (**controls**).

Proportionally less immunized among cases than among controls

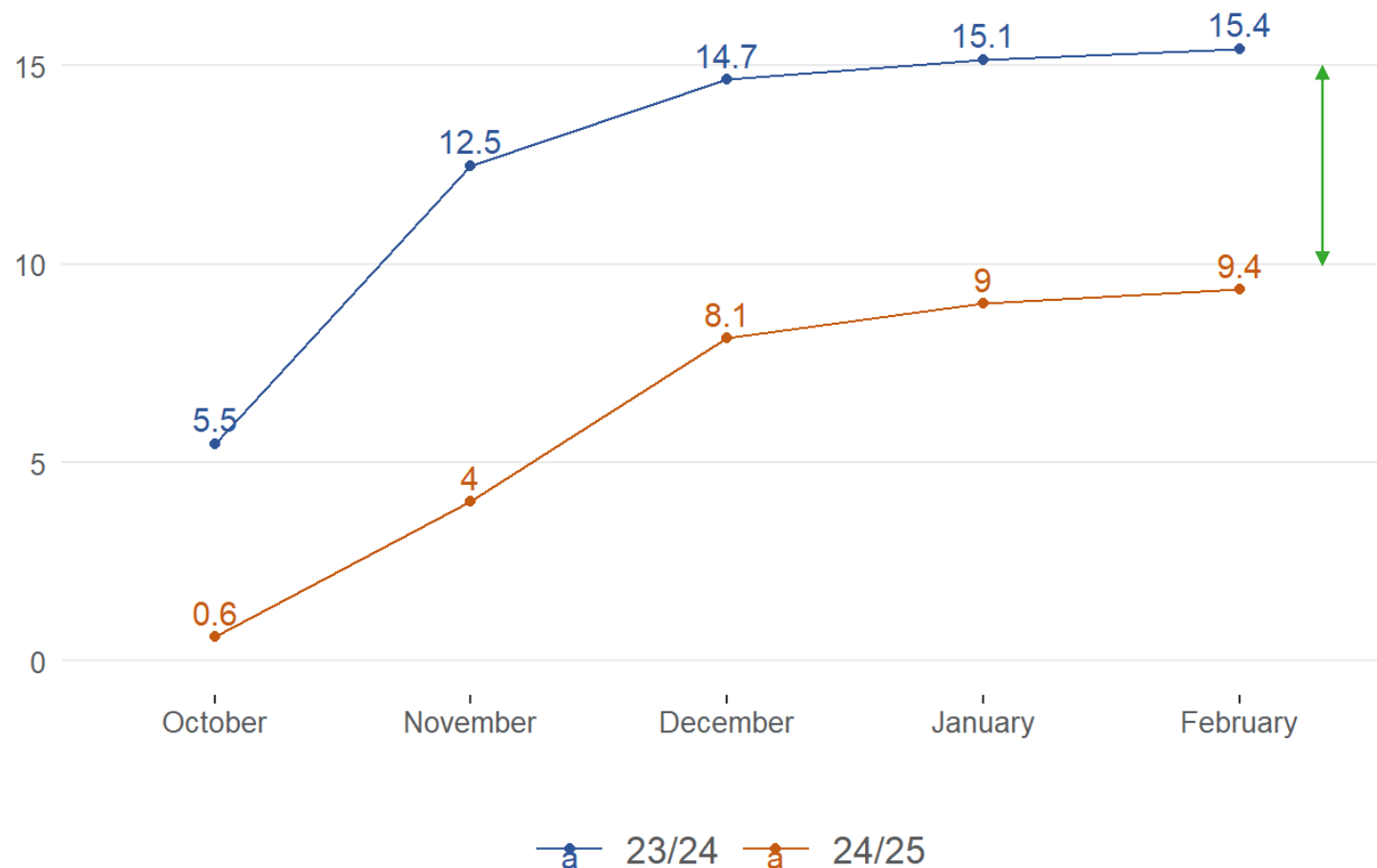
	Treated with nirsevimab	Not Treated with nirsevimab
Cases	44	142
Control	160	59

Immunisation effectiveness:

87%

Evolution in the observed incidence

Cumulative incidence for 1000 children 0-4 y.o hospitalized for RSV



Observed decrease:

40%

Avoided hospitalizations in 24-25:

~ 3600

if 90% coverage:

Additional reduction of

~ 12%

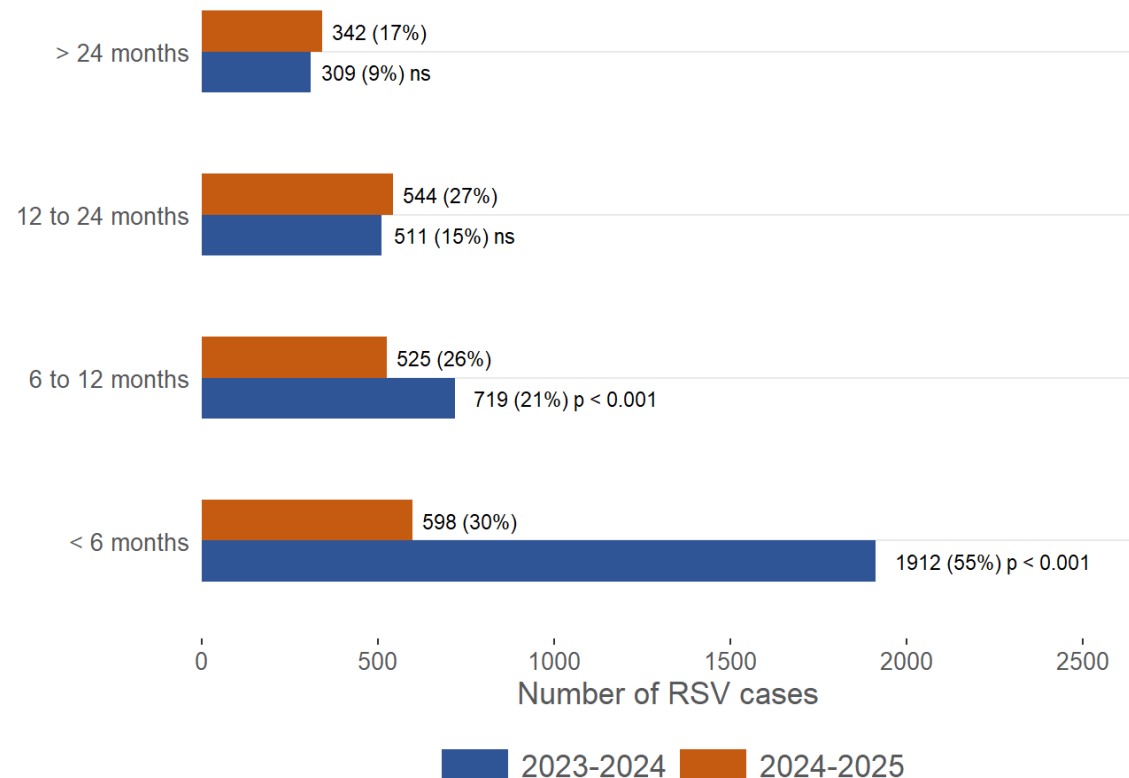
Evolution of the age distribution

Decrease of burden in <6 months in 2024-2025

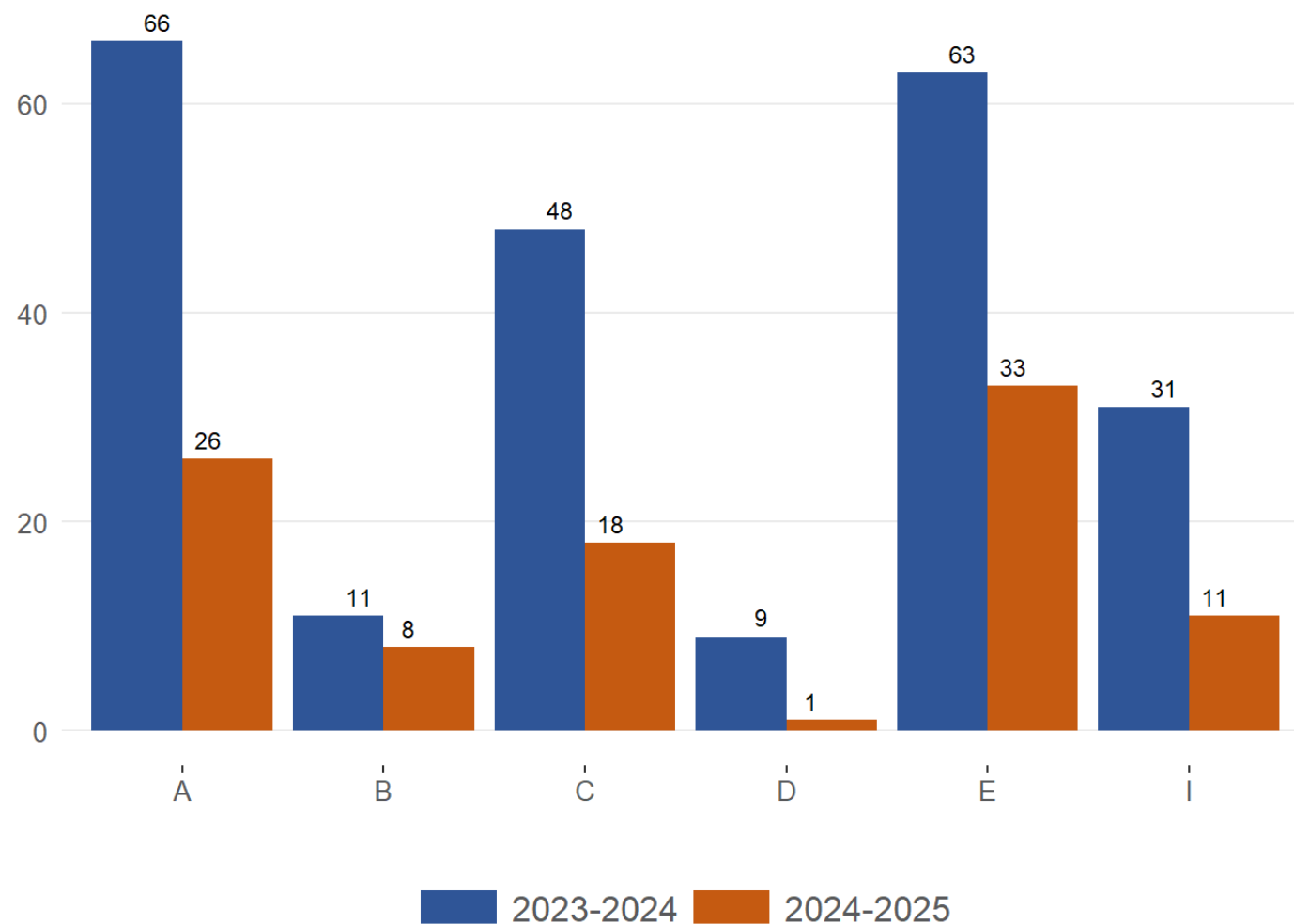
- Lower number of cases
- Lower proportion

No changes in other age groups

- No age shift



Evolution of the Pediatric ICU admission



6 centers out of 10 in Belgium
Decrease of burden in PICU admission
Observable in all PICU centers

Evolution of clinical pattern (by season)

For season 24-25 :

Higher proportion of children with comorbidities
(15% vs 20%, $p < 0,05$)

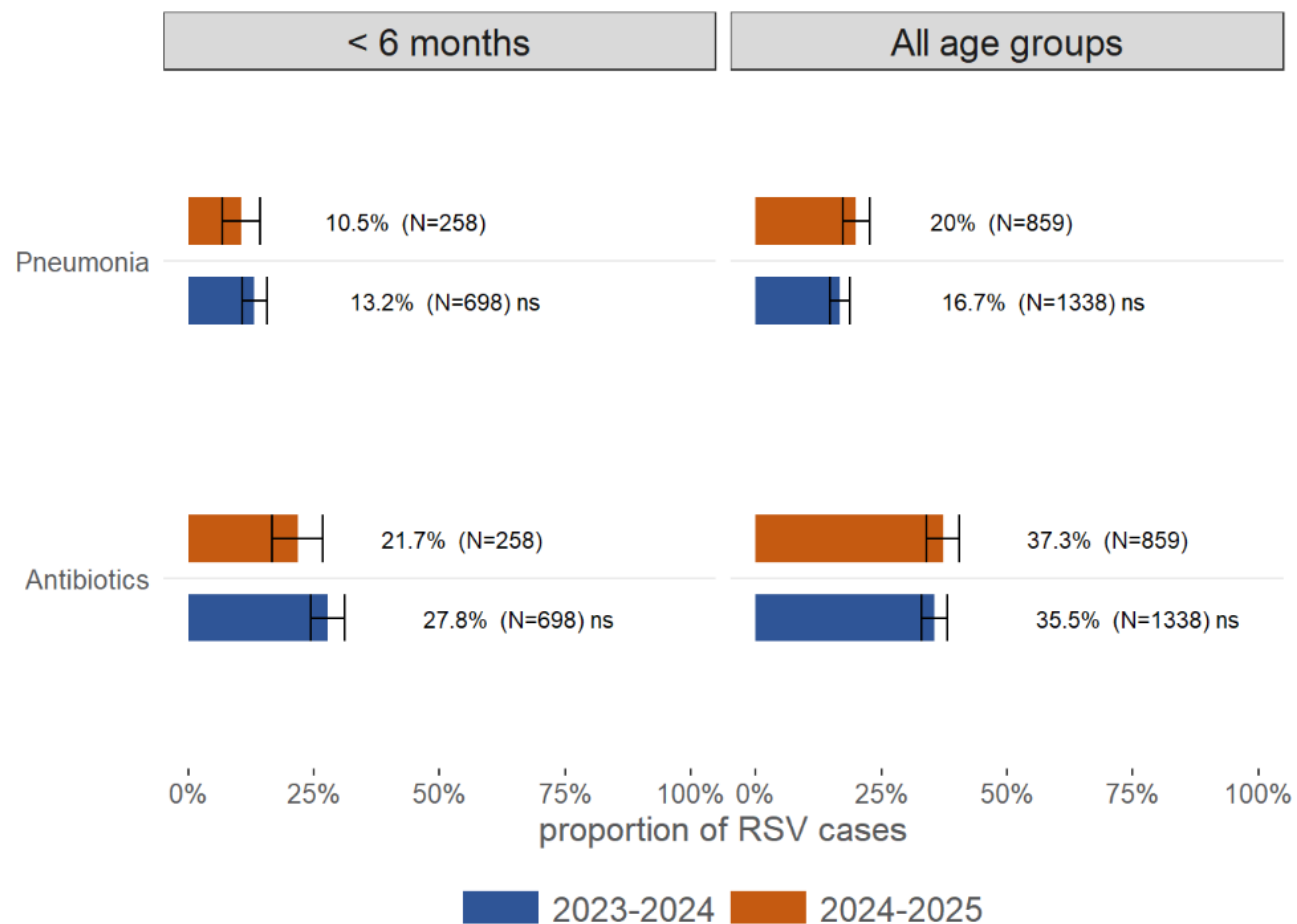
Median length of stay significantly reduced by 1 day
(5 vs 4, $p < 0,001$)

Lower proportion of children needing respiratory support
(64% vs 55%, $p < 0,001$)

Lower proportion of children needing nutritional support
(33% vs 23%, $p < 0,001$)

save of health care resources

Pneumonia and use of antibiotics (2)

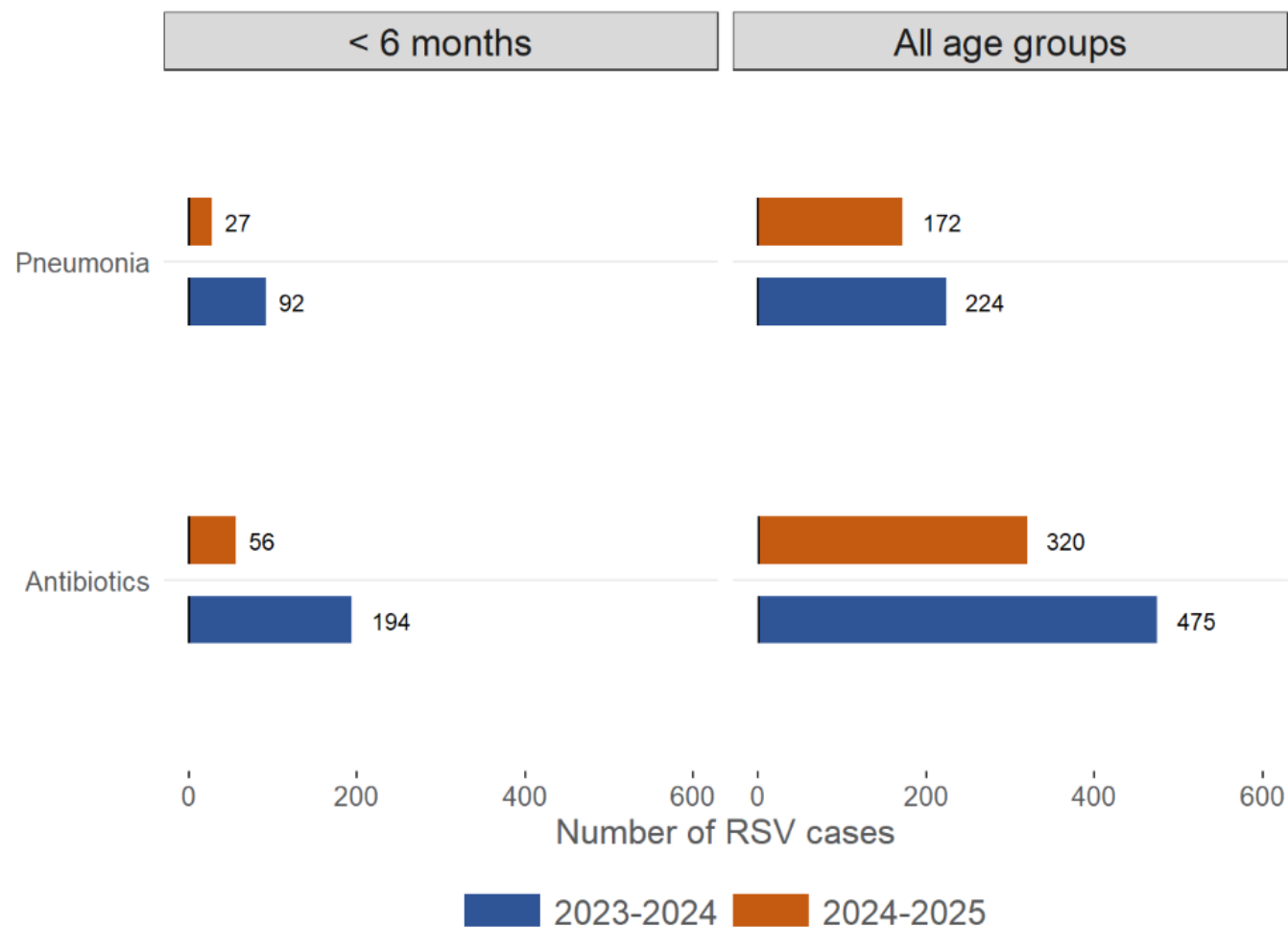


No statistical differences in the

Proportion of children presenting with pneumonia

Proportion of children receiving antibiotics

Pneumonia and use of antibiotics (2)



Reduction in

Number of children presenting **pneumonia**
Antibiotics consumption

Especially in < 6 months children

Probably due to the global decrease of
burden

Important in this era of **antibiotics resistance**

Clinical pattern according to immunisation status

	Immunized	Not immunized	
<u>Part 1: data from 20 centers</u>			
N of admissions	131	298	
Median age (IQR)	1.0 (1.0 - 3.0)	2.0 (1.0 - 5.0)	p < 0.01
Median length of stay (IQR)	4.0 (3.0 - 5.0)	4.0 (3.0 - 6.0)	ns
N(%) children needing respiratory support	60 (46%)	187 (63%)	p < 0.01

Length of stay **did not differ significantly** between both groups

Children were **younger** and needed **less respiratory support** in immunized group

Clinical pattern according to immunisation status

	Immunized	Not immunized	
<u>Part 2: Data from 15 centers</u>			
N of admissions	99	175	
N(%) children needing nutritional support	29 (29%)	60 (34%)	ns
N(%) children treated by antibiotics	21 (21%)	45 (26%)	ns
N(%) children with pneumonia	12 (12%)	21 (12%)	ns

Rate of pneumonia, antibiotics use and need for nutritional support
did not differ significantly between both groups

Lack of statistical power

CONCLUSION

Take home messages

Sub-optimal uptake of the preventive tool

High immunization effectiveness

40% of reduction of RSV pediatric hospitalizations in 2024-2025 (first season with RSV prevention) compared to 2023-2024

Highest impact seen among the **< 6 months old** (significant decrease of number of cases and in proportion)

Reduction of RSV PICU burden

Less respiratory support needed in 24-25 and among immunized children

THANK YOU!

- The coordination team of RSVPed: Sophie Blumental, Marc Raes
- Belgian Pediatric RSV Study Group (RSVPED)
- The BELSARI-NET Research Group (SARI)
- The team surveillance of respiratory infections
 - Mathil Vandromme, Yves Lafort, Sven Hanoteaux, Izaak Van Evercooren, Yinthe Dockx, Laurane De Mot
- Nathalie Bossuyt, Koen Blot (Sciensano)
- NRC respi (Sarah Denayer, Anna Parys, François Dufrasne)
- Danielle Strens (Realidad)
- Hilde Klechtermans (Jessa Ziekenhuis)
- The Care Foundation Chirec

Reasons for the limited uptake in the first year of prevention

- Heavy administrative workload linked to the reimbursement procedure for monoclonals,
- the absence of extra resources or means for public communication,
- the short delay between the decision for reimbursement and the start of the season leaving limited time to organize the catch-up campaign, the short period allowing catch up immunization,
- Moreover infants without health insurance were not eligible to receive nirsevimab.

To tackle all these hurdles,

- A national RSV task force was set up,
- initiated by the Belgian pediatricians themselves, bringing stakeholders from various institutions and healthcare structures together on a weekly basis.
- Nonetheless, the importance of a proactive educational campaign preceding implementation of a new immunization program, even in a country with a quite low rate of vaccine hesitancy, has been well demonstrated, and illustrated by the successful approach of Galician colleagues who were able to achieve a very high uptake (>90%) after 3 weeks of nirsevimab hospital-based administration program [10].

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- Because of the increase in vaccine hesitancy in Europe after the COVID-19 pandemic
- efforts should be made to improve global acceptability to prevention by parents and health care workers.
 - A study about acceptability of RSV prevention showed that parental awareness of RSV burden was low
 - but their acceptability to prevention increased after being better informed [24].
 - A quarter of them reported having lost confidence in vaccination globally after the pandemic.
 - Moreover, gynecologists were found more inclined than midwives to follow the national recommendations of prevention, whatever vaccine concerned.
 - A similar study is currently ongoing to compare results after one year of prevention and better identify strength and caveats of the first campaign, as well as parents and HCW s expectancies.

Case definition

A proven RSV related hospitalized case was defined as a child:

- ✓ presenting with a respiratory illness with **an acute onset**
- ✓ **requiring hospitalization** > 24h for this specific reason.
- ✓ **presenting with ≥ 2 following signs:**
 - Cough
 - Signs of respiratory distress
 - Fever $\geq 38^{\circ}\text{C}$
 - Abnormal pulmonary auscultation
 - Apnea/cyanosis
- ✓ in which **RSV was identified in the respiratory tract** by rapid antigen detection or PCR from nasopharyngeal swabs, nasopharyngeal aspirates or bronchoalveolar lavage within 48h after admission.